VIGNANA BHARATHI INSTITUTE OF TECHNOLOGY

(A UGC Autonomous Institution, Approved by AICTE, Accredited by NBA & NAAC - A Grade, Affiliated to JNTUH)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

PROJECT DESIGN REVIEW & LITERATURE SURVEY PRESENTATION

1. LITERATURE REVIEW & COMPARATIVE SURVEY

Paper Reference	Algorithm / Methodology	Key Advantages	Critical Limitations
Paper 1: Pratheeba et al. (IJARCCE 2025)	Survey of CSP/GA/RL/DL; Django + React.js + FullCalendar.js	Role-based access, drag-and-drop UI, mobile-friendly	No experimental validation; multi-campus scale unproven
Paper 2: Padshetty et al. (JSRT 2025)	Client-Side Rule-Based App (HTML/CSS/JS + Browser JSON)	Lightweight, serverless, personalized views	Browser storage limits scale; no AI optimization
Paper 3: Thakare et al. (IRJAEH 2026)	Hybrid Genetic Algorithm + CSP + Google OR-Tools	0% conflict rate, ~80% less effort, high accuracy	High complexity for large datasets; no real-time substitution

2. DESIGN — ARCHITECTURE (Numbered 1-10 Flow & Layers)

A. Layered Structure:

- Layer 1 (Presentation): React 19 + Zustand UI (Admin, Faculty, Student Portals).
- Layer 2 (Application Engines): CSP Timetable Solver, DSatur Exam Scheduler, Seating Matrix, Invigilation Lock, PDF Exporter.
- Layer 3 (Data Services): Firebase Auth RBAC with Student Provisioning Guard & Cloud Firestore NoSQL Database.

3. DESIGN — ALGORITHMS (Existing vs Proposed Bullet Breakdown)

Existing Algorithms (Traditional Methods & Limitations):

- Class Timetabling: Manual Trial-and-Error Grid Allocation (3 weeks effort, frequent double-booking).
- Exam Scheduling: Sequential Day-Wise Mapping (Ignores backlog course collisions).
- Exam Seating: Sequential Single-Branch Seating (High malpractice risk, over-capacity beyond 24 benches).
- Invigilation Duty: Name-Blind Round-Robin Assignment (Fails on email aliases; assigns 2 duties to 1 person).

Proposed Algorithms & AI Formulations (Our System Solution):

- CSP Solver + MRV Heuristic: Solves class timetables < 5s; locks continuous lab blocks (R25 2-hr / R22 3-hr).
- Global Cross-Dept Slot Lock: Prevents shared teacher clashes across CSE-DS, CSE-AIML, ECE, MBA & all 4 years.
- DSatur Graph Coloring Algorithm: Minimizes chromatic time slots for zero exam clashes.
- Anti-Malpractice Column-Interleaved Matrix: 4 Cols x 6 Rows (24 Benches = 48 Seats Dual / 24 Seats Single).
- Initial-Sensitive Alias-Aware Allocator: Normalizes y.raju vs v.raju & links b.amrutha.aiml@vbithyd.ac.in (max 1 duty/slot).

4. DESIGN — DATASETS (Primary / Secondary & Sample Tables)

A. Primary Data: Official VBIT Autonomous Regulations (R22/R25), 96 Multi-Branch Student Roster, 24-Bench Room Registry, Faculty Registry.

Column 0 (CSE-DS)	Column 1 (ECE)	Column 2 (CSE-DS)	Column 3 (ECE)
Row 1: 22F61A6701	Row 1: 22F61A0401	Row 1: 22F61A6707	Row 1: 22F61A0407
Row 2: 22F61A6702	Row 2: 22F61A0402	Row 2: 22F61A6708	Row 2: 22F61A0408
Row 3: 22F61A6703	Row 3: 22F61A0403	Row 3: 22F61A6709	Row 3: 22F61A0409
Row 4: 22F61A6704	Row 4: 22F61A0404	Row 4: 22F61A6710	Row 4: 22F61A0410
Row 5: 22F61A6705	Row 5: 22F61A0405	Row 5: 22F61A6711	Row 5: 22F61A0411
Row 6: 22F61A6706	Row 6: 22F61A0406	Row 6: 22F61A6712	Row 6: 22F61A0412